ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

My notebook results find 594 documents in the sci.med category.

Isolating one category is important because frequent pattern mining discovers co-occurrences within the selected transaction database, so mixing unrelated topics can dilute support counts and produce patterns that reflect cross-topic noise rather than domain-specific structure. Constraining the analysis to sci.med makes the resulting patterns more interpretable and relevant to the medical discussion domain.

AI used for this answer: Yes

If yes, how: concept explanation and notebook debugging.

External references (links):
https://scikitlearn.org/stable/datasets/real_world.html#the-20-newsgroups-text-dataset

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Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

Dropping 40% of the documents with dropna() would be flawed because it would remove a large portion of the corpus and could bias the remaining data. The missing category labels may not be randomly distributed, so the retained documents may no longer represent the original topic proportions, vocabulary structure, or document-length patterns. Also, the text of the unlabeled documents may still be useful for mining, classification, or pattern discovery, so deleting the whole row is unnecessarily destructive. In this sense, using dropna() is draconian because it discards usable observations merely because one label field is missing. One objective alternative is to train a supervised text classifier on the documents whose category_name values are known, using features such

as TF-IDF or count vectors, and then predict the missing category labels. This keeps the documents in the dataset and allows their text to contribute to the analysis rather than throwing them away

AI used for this answer: Yes

If yes, how: concept explanation and answer organization

External references (links):

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

No, the exact category proportions do not perfectly match the original. My notebook shows original proportions of approximately 0.2127 for alt.atheism, 0.2588 for comp.graphics, 0.2632 for sci.med, and 0.2654 for soc.religion.christian, while the sampled proportions are approximately 0.214, 0.278, 0.260, and 0.248 respectively. The largest differences occur for comp.graphics and soc.religion.christian, which shows that simple random sampling introduces sampling variability. The specific risk on a highly imbalanced dataset is that minority categories may become underrepresented, while majority categories dominate the sample. This can distort downstream data mining results, such as frequent pattern support counts, cluster structure, or classifier performance. A safer alternative is stratified sampling, which preserves the original category distribution in the sample.

AI used for this answer: Yes

If yes, how: concept explanation and notebook debugging; I used the notebook's sampling comparison output.

External references (links):

https://scikitlearn.org/stable/modules/generated/sklearn.model_selection.StratifiedShuffleSplit.html

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Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

After changing stop_words from 'english' to None, the total number of features increases.

My notebook shows that the document-term matrix grows from 35,482 columns to 35,788 columns, an increase of 306 features, because common English stopwords are added back into the vocabulary. Adding stopwords back reduces the overall sparsity percentage. The sparsity decreases from about 99.68% to about 99.55%, meaning the matrix becomes slightly less sparse. This is expected because stopwords appear in many documents, so their columns contain more nonzero entries than rare terms. Although the total number of zero entries may increase because the matrix has more columns, the percentage of zeros decreases, showing that adding stopwords increases dimensionality while adding mostly low-information high-frequency features.

AI used for this answer: Yes

If yes, how: concept explanation and notebook debugging.

External references (links):

https://scikitlearn.org/stable/modules/generated/sklearn.feature_extraction.text.CountVectorizer.html

https://docs.scipy.org/doc/scipy/reference/generated/scipy.sparse.csr_matrix.html

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Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

As the variance threshold increases from 0.0 to 0.05, the vocabulary size decreases sharply from 35,482 terms to 5,089 terms and then to 1,491 terms. Terms with near-zero variance are usually rare words that appear in only a few documents, or words whose counts are almost constant across the corpus. In a sparse document-term matrix, these features often contain mostly zeros and provide little ability to distinguish one document from another.

Removing them is considered safe because they add dimensionality and noise while contributing very little predictive or descriptive signal, so the reduced feature space is smaller and often just as effective for downstream mining

AI used for this answer: Yes

If yes, how: concept explanation and notebook debugging; I ran variance_filter with the three thresholds.

External references (links):

https://scikitlearn.org/stable/modules/generated/sklearn.feature_selection.VarianceThreshold.html

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

After running the correlation filter against the binary indicator for comp.graphics, the top 10 Pearson terms were thanks, keywords, looking, vga, video, card, hi, host, algorithm, and nntp. The top 10 Spearman terms were graphics, files, image, 3d, program, thanks, file, windows, ftp, and format. These are not generic English stopwords such as the, is, and, or of. However, the Pearson list includes some general Usenet or forum-style terms

such as thanks, looking, hi, host, and nntp, while the Spearman list contains more clearly graphics- and file-related terminology such as graphics, image, 3d, program, file, windows, ftp, and format.

Fundamentally, Pearson correlation measures the linear association between the raw term-count vector and the binary comp.graphics indicator, while Spearman rank correlation measures the monotonic association between their ranks. Pearson uses the actual count values and is therefore sensitive to extreme count magnitudes, whereas Spearman uses ordering and is more robust to outliers.

A word could have a high Spearman correlation but a lower Pearson correlation if its counts are consistently higher in comp.graphics documents, but a few documents contain unusually large counts for that word. Those extreme values can weaken the raw linear relationship measured by Pearson while leaving the rank-based relationship measured by Spearman relatively strong.

AI used for this answer: Yes

If yes, how: concept explanation, notebook debugging, and answer organization

External references (links):

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.pearsonr.html

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.spearmanr.html

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup: For Top-K FAE, increasing k from 100 to 500 increased the requested output size and runtime. In my run, k=100 produced 100 patterns in about 0.137674 seconds, while k=500 produced 500 patterns in about 0.487726 seconds. For MaxFPGrowth, I used a repaired/reduced transaction database because the initial generated file had a separator/format issue. In that run, MaxFPGrowth with minSup=9 took about 0.005016 seconds and returned 1 maximal pattern, while minSup=3 took about 0.005372 seconds and also returned 1 maximal pattern. The runtime increased slightly, but the pattern count did not increase because the reduced database yielded only one maximal pattern that survived the support

cut. In a fuller transaction database, lowering minSup normally expands the search space and increases the number of discovered patterns.

Answer (2) — Top-K k vs MaxFPGrowth minSup: The parameter k in Top-K FAE directly controls the number of patterns returned, aiming to keep the top k patterns according to the algorithm's ranking criterion. In contrast, minSup in MaxFPGrowth is an absolute support cutoff: it keeps maximal patterns whose support is at least minSup. Therefore, Top-K FAE gives a bounded output size, while MaxFPGrowth gives a data-dependent output size that can grow rapidly when minSup is lowered.

Answer (3) — Maximal vs standard frequent patterns: Maximal frequent patterns are more concise because they remove redundant subset patterns. If a larger itemset is frequent, many smaller subsets may also be frequent, but those smaller subsets often add little new information. Maximal mining keeps only frequent itemsets that are not properly contained in another frequent itemset, giving a compressed representation of the frequent co-occurrence structure.

AI used for this answer: Yes

If yes, how: notebook debugging, PAMI import troubleshooting, concept explanation, and answer organization

External references (links):

https://www.pami-solution.com/

https://github.com/philippe-fournier-viger/PAMI

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

The raw X_counts matrix cannot be plotted directly because it is extremely high-dimensional. In my notebook, X_counts has shape (2257, 35482), meaning each document is represented as a vector with 35,482 word-count features. A normal scatter plot can only display two or three coordinates per point, so it cannot faithfully represent documents in 35,482-dimensional space. Selecting only two columns would also discard most of the information and may not reveal the true grouping structure of the categories. Fundamentally, we must first apply a dimensionality reduction method, such as PCA, t-SNE, or UMAP, to compress the high-dimensional document-term matrix into 2D or 3D

coordinates while preserving useful similarity or cluster structure. Therefore, dimensionality reduction is required before any meaningful scatter-plot visualization of the raw text matrix can be performed.

AI used for this answer: Yes

If yes, how: concept explanation and answer organization

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html

https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html

https://umap-learn.readthedocs.io/en/latest/

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

In my notebook, the PCA plot showed the four categories as more overlapping clouds, with weaker separation around the center. The t-SNE and UMAP plots showed tighter local clusters and clearer separated groupings for the document categories. PCA struggles to separate high-dimensional text categories because it is a linear method that tries to preserve the directions of maximum global variance. In text data, the largest variance directions may be driven by common words, document length, or broad lexical frequency rather than by the nonlinear boundaries between topics. t-SNE and UMAP are nonlinear embedding methods that emphasize local neighborhood structure, so documents with similar word patterns are placed closer together in the low-dimensional plot. However, analysts must be careful not to over-interpret the physical distance between separate clusters in a t-SNE plot because t-SNE can distort global structure. Two clusters may appear far apart in the 2D visualization even if their true high-dimensional relationship is not that distant, and cluster sizes or densities should not be interpreted too literally.

AI used for this answer: Yes

If yes, how: concept explanation and answer organization

External references (links):

https://lvdmaaten.github.io/tsne/

https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html

https://umap-learn.readthedocs.io/en/latest/

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

The binarized label matrix creates 4 binary columns, one for each of the four newsgroup categories: alt.atheism, comp.graphics, sci.med, and soc.religion.christian.

Fundamentally, giving an algorithm the raw integer column 0, 1, 2, 3 creates the false mathematical assumption that the categories have a meaningful order and equal numeric spacing. For example, the model may interpret 3 as greater than 2, 2 as greater than 1, and the distance from 0 to 1 as the same as the distance from 2 to 3, even though newsgroup categories are nominal labels and not ranked quantities.

One-hot binarization eliminates this false assumption by representing each category with its own separate 0/1 indicator column. A document belonging to sci.med is represented as a binary vector with a 1 in the sci.med column and 0s elsewhere, so the algorithm sees category membership as a set of independent binary features rather than as an ordered numeric scale.

AI used for this answer: Yes

If yes, how: concept explanation and answer organization

External references (links):

https://scikitlearn.org/stable/modules/generated/sklearn.preprocessing.LabelBinarizer.html

https://scikitlearn.org/stable/modules/generated/sklearn.preprocessing.OneHotEncoder.html

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